Mini-project 2

1. **Bayesian Inverse Problem.** Consider the 1D heat problem on $(0,\pi)$ for $T = 0.01$ modeled by the operator K with observations only on the interval $(\pi/4, 3\pi/4)$. Take $x(s) = \exp(s)\sin(4s)$ and $y(t) = Kx(t)$. Let $N = 128$, define equidistant grids of s,t, and define by $Ax = y$ the discrete approximation of $Kx(t) = y(t)$. (From the matrix $A \in \mathbb{R}^{N\times N}$ for the full data problem, keep only the rows corresponding to grid points $t_i \in (\pi/4, 3\pi/4)$; this gives $A \in \mathbb{R}^{M\times N}$ with $M \approx N/2$.) Consider the inverse problem

$$Ax = y$$

in the Bayesian setting with $Y = Ax + E$. We assume the prior and noise model

$$X \sim \mathcal{N}(0, \gamma^2 I),$$
$$E \sim \mathcal{N}(0, \sigma^2 I),$$

with E independent of X and $\gamma = 1, \sigma = 10^{-2}$, so that the likelihood is $Y \,|\, X = x \sim \mathcal{N}(Ax, \sigma^2 I)$. Define the observation $y_{\text{obs}} = y + e$, where e is a realization of E.

Bayes formula gives

$$\pi_{\text{post}}(x) = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y_{\text{obs}})} \propto \exp\left(-\frac{1}{2}(x-\bar{x})^T \Gamma_{\text{post}}^{-1}(x-\bar{x})\right),$$

with

$$\Gamma_{\text{post}} = \gamma^2 I - \gamma^4 A^T(\gamma^2 AA^T + \sigma^2 I)^{-1}A,$$
$$\bar{x} = \gamma^2 A^T(\gamma^2 AA^T + \sigma^2 I)^{-1} y_{\text{obs}}.$$

   (a) Draw, say 1000, samples from π_{post}. Compute the sample mean and sample variance. Plot the sample mean, the posterior mean $\bar{x}$, the true solution x, and pointwise 95% credibility intervals $\bar{x}_i \pm 1.96\sqrt{(\Gamma_{\text{post}})_{ii}}$. Compare the sample variance with the exact posterior variance $\text{diag}(\Gamma_{\text{post}})$.

   (b) Plot the posterior standard deviation $\sqrt{(\Gamma_{\text{post}})_{ii}}$ across the interval $(0,\pi)$ and note how the uncertainty varies. Explain the phenomena. Does the true solution lie within the credibility intervals? Why (not)?

2. **Generalized Tikhonov regularization.** Consider the Abel problem for $x(s) = s^{3/4} - 1$. Discretize K by a matrix $A \in \mathbb{R}^{n\times n}, n = 1024$, and discretize similarly $x(s)$ and $y(t) = Kx(t)$ by $x, y \in \mathbb{R}^n$. Add Gaussian noise e with relative magnitude 10%, i.e. $\|e\|_2 = 0.1\|y\|_2$, to get $y^\delta = y + e$. Denote by $D \in \mathbb{R}^{(n-1)\times n}$ the forward difference matrix approximating $Dx \approx x'(s)$. (Make sure to scale with n.)

   (a) Since $x \in H^1(0,1)$ it makes sense to apply H^1 regularization in the inverse problem. We approximate the H^1-norm by $\|x\|_2^2 + \|Dx\|_2^2$, with D being the finite difference

first derivative matrix.
Show that the problem

$$x_\alpha = \arg\min_x \|Ax - y^\delta\|_2^2 + \alpha(\|x\|_2^2 + \|Dx\|_2^2) \tag{1}$$

has a unique solution for any $\alpha > 0$.

(b) Consider instead the problem with only the derivative penalty,

$$x_\alpha = \arg\min_x \|Ax - y^\delta\|_2^2 + \alpha\|Dx\|_2^2. \tag{2}$$

Recall that (2) has a unique solution for every $\alpha > 0$ if and only if $\mathcal{N}(A) \cap \mathcal{N}(D) = \{0\}$. What is $\mathcal{N}(D)$, and is the condition satisfied for the Abel matrix A?

(c) Find the solution to the ordinary Tikhonov problem (without D) for a range of α's. Plot the relative error $\|x_\alpha - x\|_2/\|x\|_2$ as a function of α on a log–log scale and identify the optimal α.

(d) Find the solution to the problem (1) with its own optimal α (found as in (c)). Compare this solution to the previous without D. Explain what you see.

3. **Large-scale deblurring of a real image with the 2D heat equation.** We now use the 2D forward heat problem from week 8 (exercise 5) on $(0,\pi)^2$ with homogeneous Dirichlet boundary conditions,

$$y(t) = Kx(t) = \int_{(0,\pi)^2} k(t,s)\,x(s)\,\mathrm{d}s,$$
$$k(t,s) = \frac{4}{\pi^2} \sum_{n,m\geq 1} e^{-(n^2+m^2)T} \sin(ns_1)\sin(ms_2)\sin(nt_1)\sin(mt_2),$$

where $x \in L^2((0,\pi)^2)$ is a grayscale image with values in $[0,1]$. Use $T = 10^{-3}$. Take a real image of size $N \times N$ with $N = 512$ as (an approximation of) x. You can use `KarinHarbo512InvertedFaded.png` or, preferably, your own photo.

On the grid of pixel centres $s_j = (j - \frac{1}{2})\pi/N$, $j = 1,\ldots,N$, the operator is discretized as

$$A = S^T \Lambda S, \qquad \Lambda = \operatorname{diag}\left(e^{-(n^2+m^2)T}\right)_{n,m=1}^{N},$$

where S is the orthonormal 2D discrete sine transform (DST-II), a linear map from $\mathbb{R}^{N\times N}$ to $\mathbb{R}^{N\times N}$. S is orthogonal, so $S^{-1} = S^T$. You may use your own implementation from week 8 or the code snippets in the hints below. Check that your code works on the basis functions: $x = \sin(ks_1)\sin(ls_2)$ (evaluated on the grid) should give $Ax = e^{-(k^2+l^2)T}x$.

(a) How much memory would it take to store A as a dense matrix? Explain why A is never formed and why an iterative method that only needs products with A and A^T is the natural choice. Show that $A^T = A$ or $K^* = K$.

(b) Load your image x (transform to grayscale, values in $[0,1]$, black is zero). Compute $y = Ax$ and add Gaussian noise e with $\|e\|_2 = 0.01\|y\|_2$ to get y^δ. Show x and y^δ.

(c) Run 200 iterations of CGLS on $Ax = y^\delta$ starting from $x^{[0]} = 0$. Plot the relative error $\|x^{[k]} - x\|_2/\|x\|_2$ against k and show the reconstructed image for a few values of k (e.g. $k = 5, 20$, the optimal k, and 200). Do you observe semi-convergence? Explain the role of the iteration number k as a regularization parameter.

(d) *(Optional)* Implement the Landweber iteration $x^{[k]} = x^{[k-1]} + \omega A^T(y^\delta - Ax^{[k-1]})$ with $\omega = 1.9/\|A\|_2^2$ (here $\|A\|_2 = e^{-2T} \approx 1$) and compare it to CGLS: the optimal k, the smallest error and the behaviour for large k.

Python snippets.

```
from PIL import Image
from scipy.fft import dstn, idstn

# image with values in [0,1], black = 0 (crop your own photo to a square first)
im = Image.open('KarinHarbo512InvertedFaded.png').convert('L')
X = np.asarray(im.resize((N,N)), float)/255

# forward operator A = S^T diag(lam) S, A^T = A
S  = lambda x: dstn(x, type=2, norm='ortho')
ST = lambda x: idstn(x, type=2, norm='ortho')
k = np.arange(1, N+1); lam = np.exp(-(k[:,None]**2 + k[None,:]**2)*T)
A  = lambda x: ST(lam*S(x))

# Landweber step (optional part)
x = x + omega*A(yd - A(x))
```

Remarks.

- CGLS: reuse your CGLS from week 8 (exercise 3). It only uses norms, so replace `K@d` by `A(d)` and `K.T@r` by `A(r)`, and start from `np.zeros_like(yd)`. Store $\|x^{[k]} - x\|_2$ and $\|Ax^{[k]} - y^\delta\|_2$ in each iteration, not the iterates.
- One product with A costs two FFT-based transforms, i.e. $\mathcal{O}(N^2 \log N)$ operations, so 200 CGLS iterations take a few seconds. Use $N = 256$ if your computer is slow.
- The data are generated with the same operator as is used for the inversion (an inverse crime).

Studying with AI tools in the mini-projects

AI tools are permitted and can be useful learning aids. Use them to support your learning and deepen your understanding of the course material.

Good practices

- Work on problems before consulting an AI tool.
- Use AI tools to explore concepts, generate alternative explanations, and test your understanding.
- Prompt for hints, ideas and feedback rather than complete solutions.
- Use AI tools for Python syntax, plotting and debugging, and make sure any retained code is understood.
- Compare and check generated output with course theory, numerical experiments and your independent reasoning; it may contain errors that look plausible.
- Generated output is often lengthy. Treat it as raw material: extract the essential ideas and present them concisely in your own words.

- Explore alternative approaches. Mathematical and computational problems can often be solved in different ways, so treat AI-generated output as one possible approach, not as an authoritative or unique answer.
- Discuss ideas with fellow students and instructors. Explaining concepts to others is one of the best ways to learn and prepare for the oral exam.
- Add a brief note in the report stating which AI tool(s) were used and for what purpose.

Rule of thumb

Make sure that you understand and can explain the reasoning and conclusions (now and at the exam) without access to AI tools. If not, the work is not yet complete.

Important notes:

- Deadline for the MP2 is Monday November 9, 2026.
- All mini-projects must be approved in order for you to qualify for the exam.
- Your background is different, and therefore you can choose to put emphasis differently in your solution.
- Demonstrate in your solution your understanding of theory and computations.
- Discuss the work with fellow students; this improves your learning and prepares for the exam.
- You may even hand in as a group; every group member is accountable for the entire solution.
- The number of pages for the main report must not exceed 7. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions. Zoom in on the relevant stuff.
- You might already now prepare the presentation of the MP to be used for the exam.